

| Backend Engineer

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📄 📄 · 📄 📄 📄 · 📄 📄 📄 📄 📄 📄 📄 📄 5📄 📄 📄 📄 📄 📄 . NestJS 📄 SaaS 📄 📄 📄 📄 · 📄 📄 · 📄 📄 · 📄 📄 · 📄 📄 📄 **Java/Kotlin/Spring** 📄 📄 📄 JVM 📄 📄 📄 📄 📄 . 📄 📄 📄 📄 ADR(Architecture Decision Record)📄 PR 📄 📄 📄 📄 📄 📄 .

📄 📄 📄 6📄 📄 📄 📄 📄 📄 SaaS 📄 📄 📄 📄 📄 , 📄 **API** 📄 **20** · 📄 📄 📄 **100** · 📄 📄 **12** 📄 📄 📄 📄 📄 📄 · 📄 📄 📄 📄 . **Kotest organization** 📄 📄 (6 PR 📄), **LINE Armeria** (1 PR 📄), **Spring Batch** · **Spring Cloud Gateway** 📄 📄 📄 📄 .

📄 📄 📄

📄	Before → After	📄
📄 📄 📄	📄 800 → 90 📄 (88.75% ↓)	📄 IP 📄 📄 , 📄 📄 📄
📄 📄 📄 📄	70% → 98%	📄 📄 📄 📄
📄 📄 (📄 6📄)	99.2%	📄 📄 📄 📄 FIFO + lease TTL
📄 📄 📄 📄	0 / 6 📄	Idempotency-Key + 📄 📄 📄
📄 📄 📄 webhook race	📄 + 📄	📄 📄 📄 📄 📄

📄 📄 = 📄 📄 / 📄 📄 📄 · 📄 📄 = JVM/Spring 📄 📄 📄 📄 📄 . 📄 📄 📄 📄 📄 📄 .

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Production — 📄 📄 📄 📄 📄 📄

- TypeScript · NestJS · Node.js · TypeORM
- MySQL / Aurora · PostgreSQL · Redis (Cluster · Lua · 📄 📄 · Token Bucket)
- RabbitMQ (Quorum Queue · DLX) · 📄 📄 📄 application-level retry
- Docker · Docker Compose · Traefik 📄 📄 · AWS ECS · 📄 📄 📄 📄
- Prometheus · Grafana · APM (span tagging) · Slack Webhook · JFR

JVM / Spring Re-design — 📄 📄 📄 📄 · 📄 📄 📄

- Java · Kotlin · Spring Boot · Spring Batch · JPA · QueryDSL
- Redisson · Resilience4j (Circuit Breaker · Retry · Bulkhead · RateLimiter · TimeLimiter)
- Kafka (Outbox · CDC · Consumer Group) · ShedLock · OAuth2

Evidence — 📄 · 📄 · 📄 📄

- Testcontainers · JUnit 5 · Kotest · Pytest · Jest · nGrinder
- Kotest organization 📄 · Armeria 📄 · Spring Batch 📄 📄

🔖 📖

JVM 📦 📦 📦 📦 — 📦 7 · 📦 📦 + **PR** 📦 📦.

[Kotest](https://github.com/kotest/kotest) — Kotlin 📦 📦 📦 📦 (4.7k stars)

2026.03 📦 📦 sksamuel 📦 📦 📦 **organization** 📦 📦, 📦 📦 6 PR 📦.

- 📦 📦 Test Metadata Public API 📦 📦 (+355/-16, 5 📦 16h 📦 📦) — [issue #5103](https://github.com/kotest/kotest/issues/5103), [PR #5905](https://github.com/kotest/kotest/pull/5905)
- 📦 📦 data class 📦 📦 diff — [issue #2545](https://github.com/kotest/kotest/issues/2545), [PR #5835](https://github.com/kotest/kotest/pull/5835)
- JSON Schema anyOf / oneOf 📦 📦 — [issue #4463](https://github.com/kotest/kotest/issues/4463), [PR #5807](https://github.com/kotest/kotest/pull/5807)
- JSON Matchers 📦 📦 Json 📦 📦 📦 — [issue #4601](https://github.com/kotest/kotest/issues/4601), [PR #5795](https://github.com/kotest/kotest/pull/5795)
- @OnlyInputTypes 📦 📦 📦 📦 — [issue #5589](https://github.com/kotest/kotest/issues/5589), [PR #5789](https://github.com/kotest/kotest/pull/5789)
- shouldHaveSingleElement 📦 📦 📦 — [issue #5755](https://github.com/kotest/kotest/issues/5755), [PR #5756](https://github.com/kotest/kotest/pull/5756)

[Armeria](https://github.com/line/armeria) — LINE 📦 Java 📦 📦 📦 RPC 📦 📦 📦 (5.1k stars)

- RequestContextExporter BuiltInProperty 📦 + Logback MDC export 📦 📦 (+339/-5, **MERGED** 2026.04) — [issue #4403](https://github.com/line/armeria/issues/4403), [PR #6683](https://github.com/line/armeria/pull/6683)

[Spring Batch](https://github.com/spring-projects/spring-batch) — Spring 📦 📦 📦 📦 📦 📦 (2.9k stars)

📦 📦 📦, PR 📦 📦.

- fault-tolerant step chunk scanning 📦 📦 4📦 📦 📦 — [issue #3946](https://github.com/spring-projects/spring-batch/issues/3946) (📦 📦 📦 "documentation issue" 📦 📦)
- conditional flow decider order 📦 — [issue #4478](https://github.com/spring-projects/spring-batch/issues/4478) (📦 📦 📦 "this is a bug" 📦)

[Spring Cloud Gateway](https://github.com/spring-cloud/spring-cloud-gateway) — Spring reactive API gateway

5.1.x 버전

- Apache HttpClient5 버그 이슈 — [issue #3311](https://github.com/spring-cloud/spring-cloud-gateway/issues/3311) (리소스 풀 + 버그)
- API Versioning Predicates 버그 이슈 — [issue #4024](https://github.com/spring-cloud/spring-cloud-gateway/issues/4024) (리소스 풀)

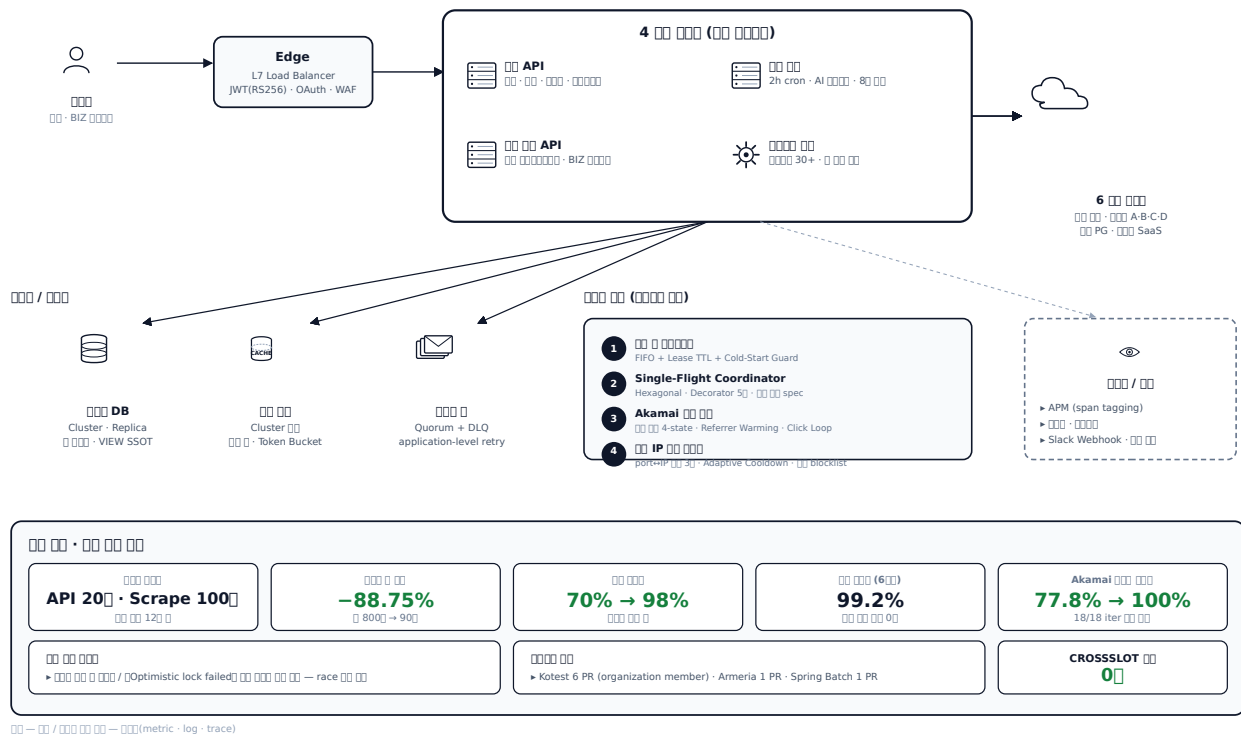
— (Lemong) | | 2024.11 ~

프로젝트

서비스는 SaaS / Biz 서비스. 4 서비스 (API, , ,)
서비스 (Cluster) · DB · 가

A1. 서비스 SaaS —

4 서비스 · 6 서비스 · 서비스



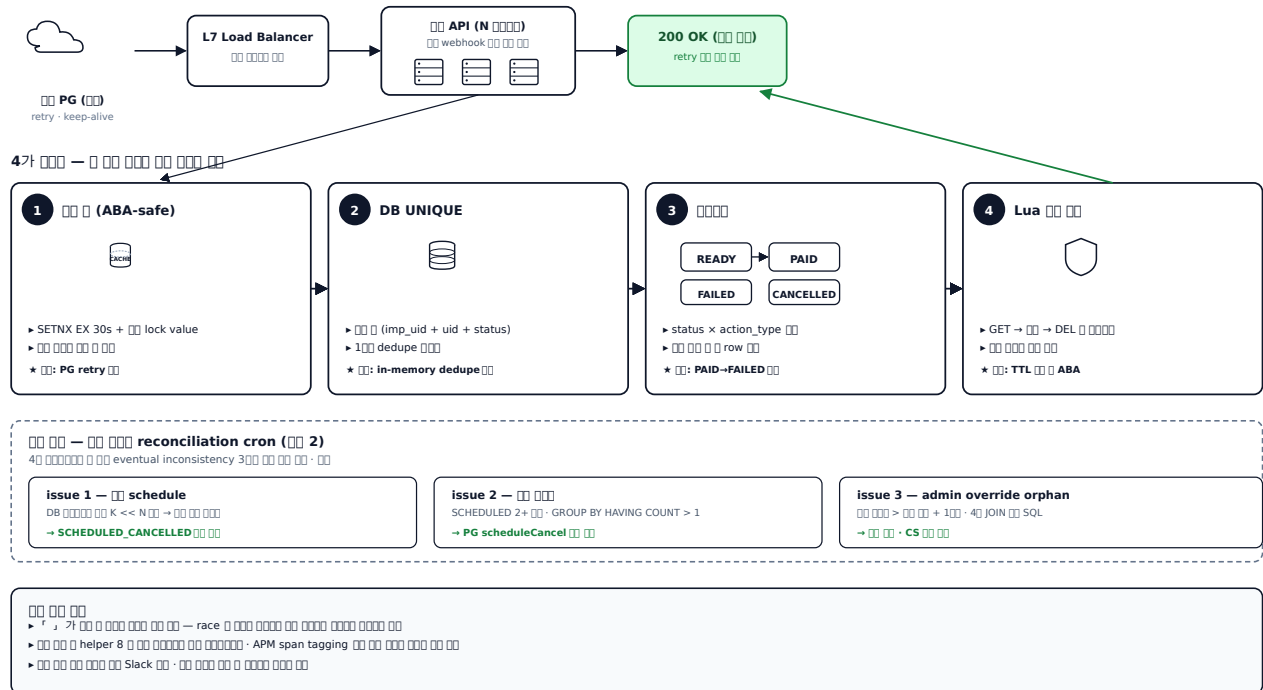
서비스

5 서비스 + 서비스 + JVM/Spring 서비스

□□ □□ + □□□ reconciliation – 5 □ □ □ □ □ □ □ □ □ □

A2. webhook 4000 — DB 4000 · 4000 · 4000

race · ABA · · out-of-order 4 hazard



webhook 4 + reconciliation

PG webhook 5.

- PG retry (5xx / timeout)
- `retry` schedule 1st `retry` fallback retry → PAID / FAILED `retry` `retry` `retry`
- `retry` `retry` `retry` · `retry` `retry` `retry` `retry`
- Load Balancer keep-alive `retry`

INSERT 1 row 00 00 → 00 00 → 0000 20 00 → 0000 00 00 → 00 0 00 00 000. 00 0000 000 in-memory dedupe 가, ORM 1 findOne → save 0000 read-modify-write 0000 race window, 00 00 0 0000 0 TTL 00 0 ABA 00 (Martin Kleppmann Redlock 0000 00), DB UNIQUE 00000 PG retry 0000 10 00 0 0.

가 webhook 400 에러를 발생시키는 원인은 PG의 DB가 source of truth가 아니며, schedule (PG 실행 + DB cancel), admin override orphan 처리 / PG 실행 실패 처리 때문이다.

`reconciliations` = `'r' / DB / YYYY / MM DD HH 4D DD + DD reconciliation cron DD DD DD.`

4. 問題 (問題 1):

1. **ABA-safe** — SETNX EX 30s + lock value. Stripe Idempotency Key lock value ownership

2. **DB UNIQUE** — $\text{imp_uid} + \text{merchant_uid} + \text{status} \cdot 1$ dedupe window · fail-open
row

3. — $\text{READY} \rightarrow \text{PAID} / \text{FAILED} / \text{CANCELLED} \times \text{action_type}$ · (PAID · FAILED) row

4. **Lua** — GET → DEL · TTL · ABA

reconciliation cron — 3 inconsistency

- **schedule** — DB $K \ll N$ PG `getSchedulesByBillingKey` → 4-way
- — `GROUP BY user_id HAVING COUNT > 1` SQL
- **admin override orphan** — `active.end_date > scheduled.schedule_at + 1month` 4 JOIN SQL

— `(findOne((user, coupon)) idempotent return)` · `(QueryRunner)` · `schedule` `amountWithUserCoupons fallback`

()

- 「 merchant_uid=... 」 race +
- SETNX + Lua DEL helper 8 (promotional upgrade · change plan · verify code ·) helper
- 09:10 KST Slack — / CS
- 0 (UNIQUE + findOne), update 0

— 4 + reconciliation cron · (Slack)

Spring — `@Transactional propagation` · `Redisson watchdog` · `JPA @Version`

`OptimisticLockException` · `ER_DUP_ENTRY` → `@RestControllerAdvice` · `ShedLock cron` ·

`QueryDSL` ()

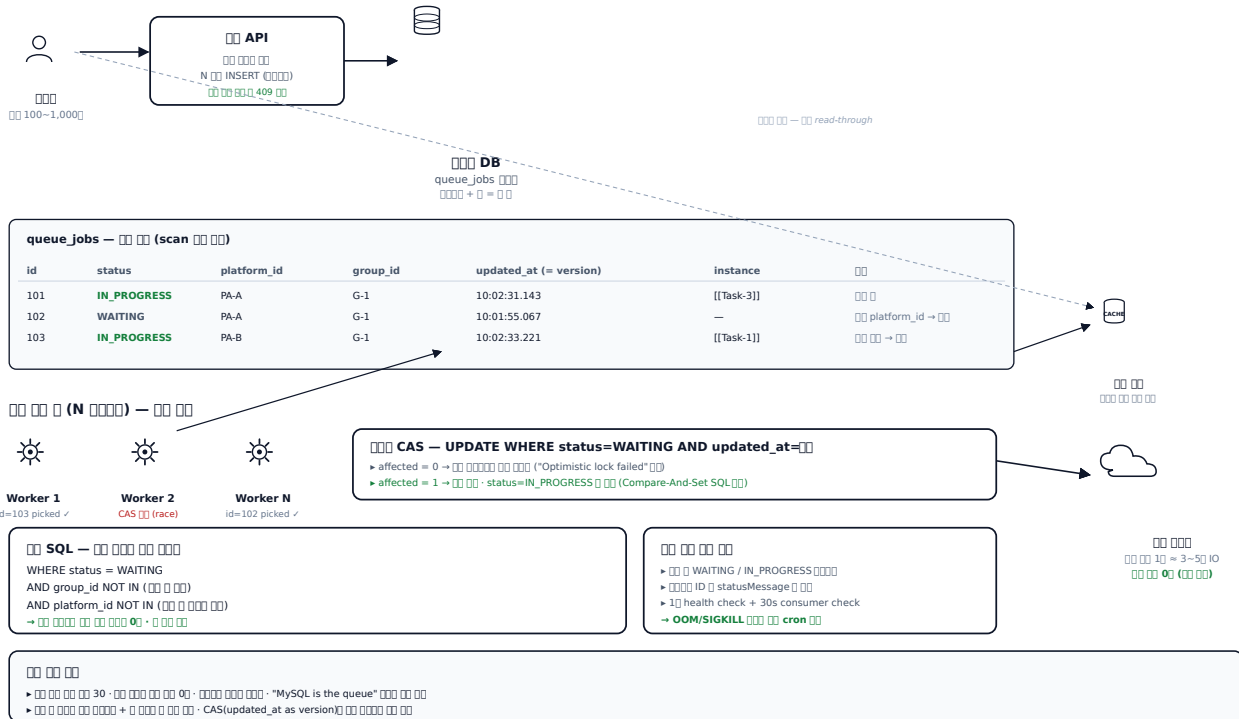
[commerce-comment-platform-be](https://github.com/PreAgile/commerce-comment-platform-be)

()

DB + CAS + RabbitMQ

B1. 如何 DB 查詢 — CAS 如何查詢

「MySQL is the queue」 · 가 CAS · ∈ 0000 00 0000 · 00 00 00 00



DB + Hierarchical

□□ — □□ □□□ □□□ □ □□ □□□ □□□□□.

- 00 000 00 00 (0 0 $\approx$ 3~5 IO) — 0000 0100~1,000」 000 000. HTTP 00 00 0 Load Balancer 50 0000 00, 00 0000 in-memory queue 0 000 0 00 00, 00 000 00 000 0000 worker 200 00 000 0 00 000 0 00 → 0 00 00 00
- 200 **cron** 00 00 — 00 000 10 00 00 **13,127** 00 / **5,719** 00 00 / **13,699 task** 00. 00 0000 00 00 0 00 00 0 captcha / 0000 / 00 000. 00 00 ThreadPool 0 00 00 000 00 00 00 00 = 00 00, 00 000 1100 → 200 cron 00 0 00

Our work is orthogonal to the work on fire-and-forget [10] and stateful [11, 12, 13] message queues. BullMQ is not a fire-and-forget message queue, nor is it a stateful message queue. BullMQ is a message queue that is designed to be used in a fire-and-forget manner, but it is also designed to be used in a stateful manner.

□□ — □ □□□ □□□□□.

(a) **DB** **CAS** — **updated_at** **version** **ORM** = **CAS** (**Compare-And-Set**) **SQL**. **affected = 0** **SQL** **platform** **id NOT IN** **ORM**.

(b) `getWorkerThread / getWorkerThread` **hierarchichal** `workerThread` — (`platform_id`, `password`) `workerThread` → `ThreadPool` `workerThread`. `workerThread` `worker thread` `workerThread` `workerThread`. `workerThread` `workerThread` `_deactivate_shops` `workerThread` `workerThread` (`workerThread`) — `workerThread` `workerThread`.

이 (이 이) — 이 이 이 이 이 이 이 이 AAS 이 이 이 이 → **RabbitMQ** 이 이 이 이 이 이 이 이 . 이 이 이 / 7 8 이 stateful 이 이 fire-and-forget 이 이 이 이 이 이 , DB 이 이 이 이 source of truth 이 이 이 이 fan-out 이 이 이 이 이 이 .

이 (이 이)

- 이 이 이 이 30, 이 이 이 이 + 이 이 이 이 이 이 이 이
- 이 이 Job N modified by another instance (optimistic lock failed) 이 이 — race 이 이 이 이 이 이 이 이
- 이 이 이 이 0, 이 이 이 이 이 이
- 이 이 이 13,127 이 2 이 cron 이 이 , 이 platform_id 이 이 0
- 이 이 이 이 이 이 · 이 이 이 이 이 (이 이)

이 이 — RDS Queue 이 / 이 SQL CAS 이 / 이 이 이 이 (이 이 이 + health check) 이 이 .

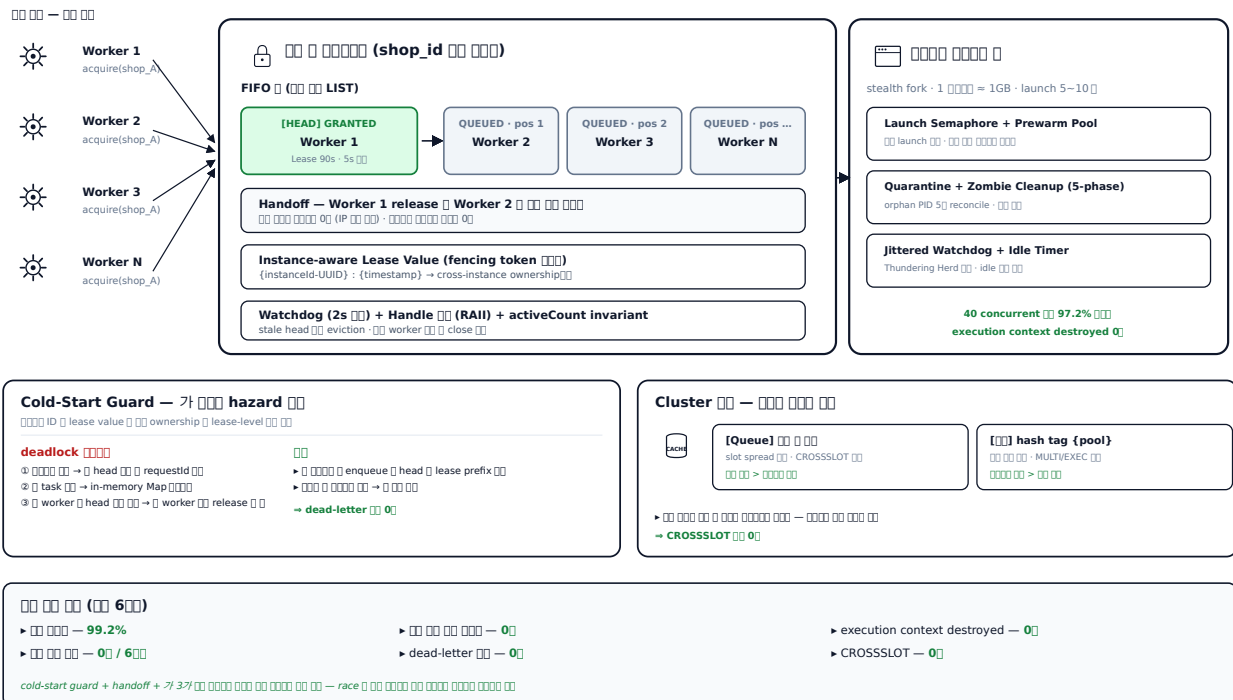
hierarchical 이 이 + 이 이 이 dict 이 . 이 RabbitMQ 이 이 이 ADR 이 이 .

Spring 이 — JPA @Version 이 + OptimisticLockException (CAS) · Spring Batch + ShedLock + QuerydslZeroOffset Reader · Resilience4j Bulkhead.semaphore (이 이 이) + CircuitBreaker (이 이 이) · Kafka Outbox / Consumer 이 + DLQ (이 이 이)
[commerce-batch-orchestrator](https://github.com/PreAgile/commerce-batch-orchestrator) 이 RabbitMQ↔Kafka 이 이)

이 이 이 + 이 이 — **30+** 이 이 이 **race** · 이 이 이 이 이

C1. 이 이 이 — FIFO + Lease TTL + Cold-Start Guard

30+ 이 이 이 race 이 4 layer 이 + Handoff 이 + 이 3



이 이 이 + Cold-Start Guard

□□ — 30+ worker □ □□ □□ □□ □□ □□ race + □□ □□□ □ □□ □□□ □□□□ □□□ □ □□ □□ □□ □□□ □□□.

□□ race:

- □□ □□ □ page □□ navigation = □□□□ □□□□ execution context destroyed
- □□ □□ □□ □□ □ □ = □□ □□ IP □□ □□ □□ (1~24□□ □ □□ □□ □□ □□)
- □□□□ □□□□ 1 □□□□ ≈ 1GB+, launch 5~10□ — race □ □□□·□□□ N□ □□
- □□ □□ Cluster □□□ □□ □□□□ □□ □ □□ □ CROSSSLOT
- 가 □□□□ hazard: **cold-start deadlock** — 가 □□□ □□□□ in-memory Map □ □□□□□ □□ □□ □ head □ □□□ requestId. □□□ worker □ □ head □□□ □□□□ □ worker □ □□□ □□□ release □ □

□□ □□□ □□□:

- □□ □□ □□□□ □ □□ □□□□ □□□ □□□□ 99% □□
- □□ □□□□ □□ □□□ pending → verified → blocked □ □□ □□□ □□□, □ □□□ 4 □□ silent □□□□ □□ □□□ password retry □ □□ □□□□

□□ — □ □□ □□□□ □□□ □□□ □□□ □□ + □□ □□□ □□□ □□□ □□.

(a) □□ □ □□□□□ — **4 layer** □□□:

- FIFO □ (□□ □□ LIST) · per-request lease (90s TTL · 5s □□)
- **Instance-aware lease value** = {instanceId-UUID} : {timestamp} (Martin Kleppmann fencing token □□ □□□ — cross-instance ownership □□□□ lease-level □□)
- Handoff pattern (release □ □□ □ entry □ □□ □□ □□□)
- □□□ 3□ — forceRelease (□□ kill switch) · forceTerminate (activeCount □□ 0) · evictStaleHead (2s □□ stale head LREM)
- □□ □□ Cluster CROSSSLOT — queue □ □□ □ □□ (□□ □□ □□), □□ □□□ hash tag {pool} (□□□□ □□ □□) □□ □□□ □□□ □□ □□

(b) □□ □□□ □□□ □□ — □□ □□ □□□:

- □□ □□ 4-state □□□□ (□□ / □□ / challenge / □□) — 「 □ □□ challenge > □□ > 」 □□□□ □ □□□ 4 □□ silent □□□ □□ □□
- □□ □□□ service □ spec □ SSOT □□ □□□ □□ (discriminated union □□)
- Referrer Warming — □□ URL □□ referrer chain □□, 15□ human-like mouse jitter □□ → sensor telemetry □□ → □□ □□ □□
- □□ □□ helper □ main loginResult / Quick Retry / Prewarm Swap 3 □□□□ □□ □□ (silent □□ □□)

□□ (□□ □□ · □□ 6□□)

- □□ □□□ **99.2%**, □□ □□ □□ □□□ **0□**
- □□ □□ □□ **0□**, execution context destroyed □□ **0□**
- □□□□ □□□ □ dead-letter □□ **0□** (cold-start guard □□)
- CROSSSLOT □□ **0□** (hash tag / □□ □ □□ □□ □□ □□)
- □□ □□□ □□□ □□□ **77.8% → 100%** (Referrer Warming □□ · 18/18 iter □□ □□ · PR □□□ iter × worker □□□ □□ □□ □□□ □ □□□□ □□□□)

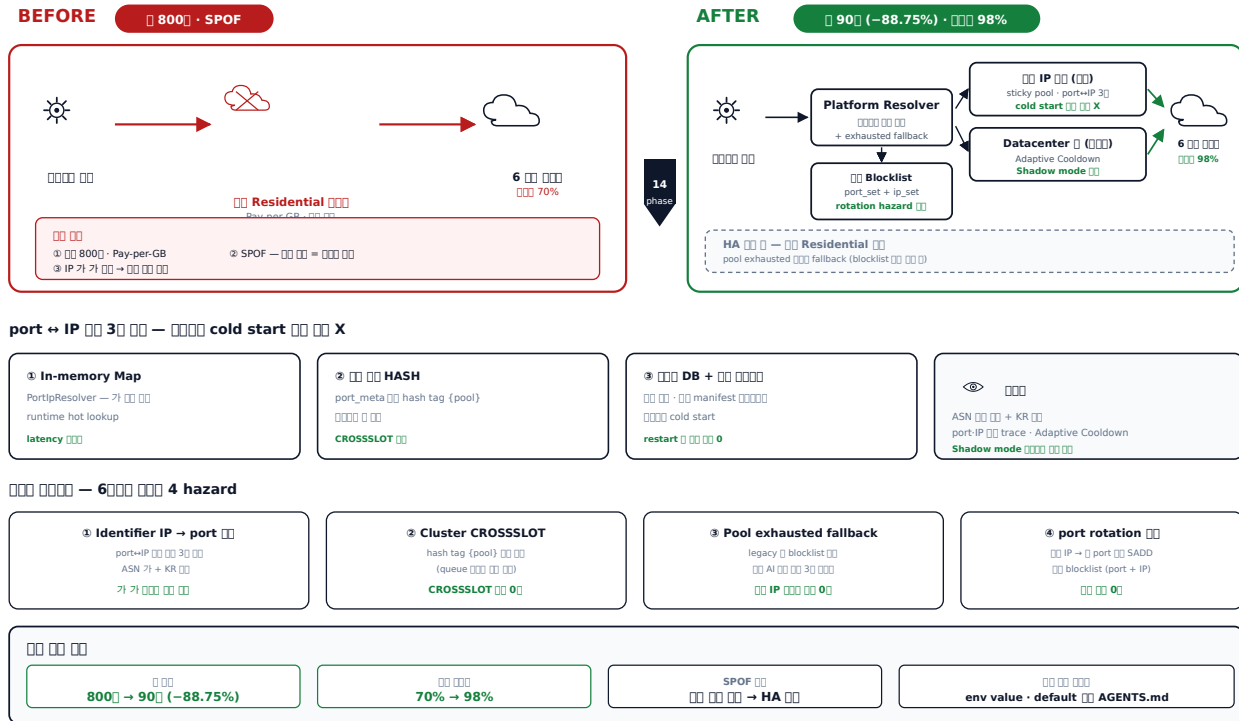
□□ □□ — □□ □ □□□□□ (4 layer + □□□ 3□) □□ · □□ · □□. □□ □□ □□□□ □□ □□ spec / service SSOT □□ □□□□.

Spring — Redisson RFairLock (FIFO + lease auto-extend) · Kotlin Coroutines Mutex.withLock per-key in-process layer · sealed class discriminated union · Resilience4j Bulkhead
 (commerce-external-gateway-kt)(https://github.com/PreAgile/commerce-external-gateway-kt) 7 spec

IP — 88.75%

C2. IP — 88.75% · 70% → 98%

14 phase phased rollout — port→IP 3 · Adaptive Cooldown · blacklist · pool exhausted fallback



IP Before / After

— Residential SPOF · 800 =

- / SPOF — 800 · =
- **Identifier** — Datascenter port IP 가 가 (IP) stable identifier (port) 가
- **Cluster CROSSLOT** — · STREAM · blacklist MULTI/EXEC
- **port rotation** — IP port mapping port blacklist

— 6 14 phase phased rollout. phase → → phase cutover.

4:

- **port→IP 3** (In-memory Map + HASH + DB + manifest) — cold start
- **blacklist** — port_set + ip_set. allocator port IP lookup → ip_set hit port port_set 가. port rotation hazard

- **Adaptive Cooldown** — 5 outcome (success / block / timeout / networkError / authError / siteChange / unknownError) × consecutiveFailures × latency health check, Shadow mode check
- **Pool exhausted → legacy fallback** — NaverIspPoolExhaustedError throw → legacy path, legacy selectPortRespectingBlocklist check IP skip

AGENT.md check — env value check · deprecated alias check default check. fallback check AI check 3 follow-up check PR check.

check (check check)

- check 800 → 90 check (88.75% check) — check check
- check 70% → 98% — check check check
- check SPOF check (HA check check), check check
- check IP check legacy check check 0
- CROSSLOT check 0 — hash tag {pool} check

check — 14 phase phased rollout check phase check ADR check · check · check · check check check check. check AI check check follow-up check.

Spring check — Spring Cloud Gateway + Resilience4j Retry / CircuitBreaker / Bulkhead check ·

WebClient ExchangeFilterFunction check proxy resolver check · @ConfigurationProperties +

@Validated check env check (check check)

[commerce-external-gateway-kt](https://github.com/PreAgile/commerce-external-gateway-kt))

BIZ 1000 – 200 00 + SQL VIEW + 00 IO 00 00 00

B3. BIZ 1000 – 2 100 000 + SQL VIEW SSOT + 10 IO 100 000 000

1000 → 1,000 × 70 = 70,000 — N+1 joins + JOIN sweet spot

[!\[\]\(deab1c35b8bdbc17e1165ce3b654c399_img.jpg\)
!\[\]\(fa69b1762d528c5ed8515289d723f7a3_img.jpg\)
!\[\]\(db3a1055ec017f877b0bc279d2ac453d_img.jpg\)
!\[\]\(0c1f7be6fe4574aea9c459ea7dcd6cbc_img.jpg\)](#)
[JOIN —](#)
[!\[\]\(c6094f254bbf0898281ec49099c8e535_img.jpg\)
!\[\]\(45a4cefa11519256d3c4ab041b1d13cb_img.jpg\)
!\[\]\(4b2715f0e01e128b1046516e89295e82_img.jpg\)
!\[\]\(5dabb9db2c5e53a211dd68beb15bff44_img.jpg\)](#)

```
SELECT ...  
FROM brands b JOIN brand_users bu ... JOIN shops s ... JOIN orders o ...  
WHERE b.brand_id = ? AND sales_date BETWEEN ?
```

$$\rightarrow 7,000 + 1 \times N \times 0.0001$$

✓ Sweet spot — 2 ☐ batch select

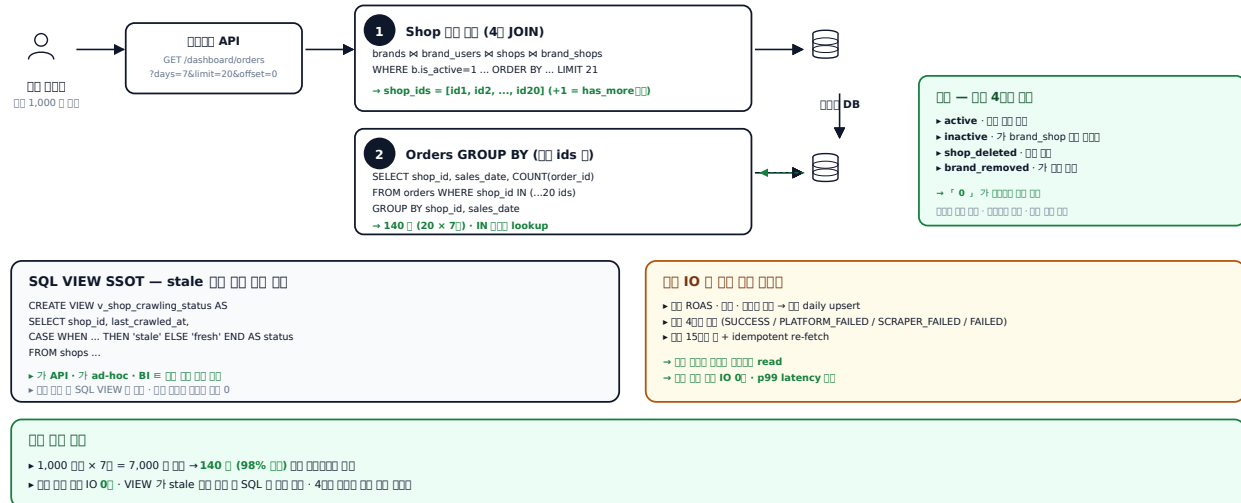
```

① Shop 000 000 (LIMIT/OFFSET DB-level)
→ 4 JOIN (brand x brand_user x shop x brand_shop)
→ ORDER BY : LIMIT %s+1 OFFSET %s (has_more=1)

② shop_ids 0 0000 orders GROUP BY
→ WHERE shop_id IN (...20 ids)
= 140 (20 x 7) = 00000 00
▶ N+1 00 + 00 JOIN orders

```

2-step batch



BIZ 2000 2000 2000 + SQL VIEW SSOT

□□ — □□□□□ □□ 1□□ □□ 1,000□□□ □ □□□□ □□□.

□ □□□ □□□ □□ □: □□□ □□ 7□ □□ □□, □□ ROAS / □□ / □□, □ □□ 7□, □□□ □□ (stale) □□, □□□ □□.

- JOIN 表 — brand → ... → orders 表 表 表 表 表 (1,000 × 7 × 表 N)
- 表 表 / 表 IO 表 表 表 表 表 表
- ORM 表 表 4 表 表 + dynamic where + order by + LIMIT 表 表 表 表

② — 「N+1 ③ ④ JOIN ⑤ sweet spot ⑥ 2 ⑦ batch select」 + 「SQL VIEW ⑧ ⑨ SSOT」 + 「IO ⑩ ⑪ ⑫ ⑬ ⑭ ⑮ ⑯ ⑰ ⑱ ⑲ ⑳ ㉑ ㉒ ㉓ ㉔ ㉕ ㉖ ㉗ ㉘ ㉙ ㉚ ㉛ ㉜ ㉝ ㉞ ㉟ ㊱ ㊲ ㊳ ㊴ ㊵ ㊶ ㊷ ㊸ ㊹ ㊺ ㊻ ㊼ ㊽ ㊾ ㊿ ㏀ ㏁ ㏂ ㏃ ㏄ ㏅ ㏆ ㏇ ㏈ ㏉ ㏊ ㏋ ㏌ ㏍ ㏎ ㏏ ㏐ ㏑ ㏒ ㏓ ㏔ ㏕ ㏖ ㏗ ㏘ ㏙ ㏚ ㏛ ㏜ ㏝ ㏞ ㏟ ㏠ ㏡ ㏢ ㏣ ㏤ ㏥ ㏦ ㏧ ㏨ ㏩ ㏪ ㏫ ㏬ ㏭ ㏮ ㏯ ㏰ ㏱ ㏲ ㏳ ㏴ ㏵ ㏶ ㏷ ㏸ ㏹ ㏺ ㏻ ㏼ ㏽ ㏾ ㏿ 㐀 㐁 㐂 㐃 㐄 㐅 㐆 㐇 㐈 㐉 㐊 㐋 㐌 㐍 㐎 㐏 㐐 㐑 㐒 㐓 㐔 㐕 㐖 㐗 㐘 㐙 㐚 㐛 㐜 㐝 㐞 㐟 㐠 㐡 㐢 㐣 㐤 㐥 㐦 㐧 㐨 㐩 㐪 㐫 㐬 㐭 㐮 㐯 㐰 㐱 㐲 㐳 㐴 㐵 㐶 㐷 㐸 㐹 㐺 㐻 㐼 㐽 㐾 㐿 㑀 㑁 㑂 㑃 㑄 㑅 㑆 㑇 㑈 㑉 㑊 㑋 㑌 㑍 㑎 㑏 㑐 㑑 㑒 㑓 㑔 㑕 㑖 㑗 㑘 㑙 㑚 㑛 㑜 㑝 㑞 㑟 㑠 㑡 㑢 㑣 㑤 㑥 㑦 㑧 㑨 㑩 㑪 㑫 㑬 㑭 㑮 㑯 㑰 㑱 㑲 㑳 㑴 㑵 㑶 㑷 㑸 㑹 㑺 㑻 㑼 㑽 㑾 㑿 㒀 㒁 㒂 㒃 㒄 㒅 㒆 㒇 㒈 㒉 㒊 㒋 㒌 㒍 㒎 㒏 㒐 㒑 㒒 㒓 㒔 㒕 㒖 㒗 㒘 㒙 㒚 㒛 㒜 㒝 㒞 㒟 㒠 㒡 㒢 㒣 㒤 㒥 㒦 㒧 㒨 㒩 㒪 㒫 㒬 㒭 㒮 㒯 㒰 㒱 㒲 㒳 㒴 㒵 㒶 㒷 㒸 㒹 㒺 㒻 㒼 㒽 㒾 㒿 㓀 㓁 㓂 㓃 㓄 㓅 㓆 㓇 㓈 㓉 㓊 㓋 㓌 㓍 㓎 㓏 㓐 㓑 㓒 㓓 㓔 㓕 㓖 㓗 㓘 㓙 㓚 㓛 㓜 㓝 㓞 㓟 㓠 㓡 㓢 㓣 㓤 㓥 㓦 㓧 㓨 㓩 㓪 㓫 㓬 㓭 㓮 㓯 㓰 㓱 㓲 㓳 㓴 㓵 㓶 㓷 㓸 㓹 㓺 㓻 㓼 㓽 㓾 㓿 㔀 㔁 㔂 㔃 㔄 㔅 㔆 㔇 㔈 㔉 㔊 㔋 㔌 㔍 㔎 㔏 㔐 㔑 㔒 㔓 㔔 㔕 㔖 㔗 㔘 㔙 㔚 㔛 㔜 㔝 㔞 㔟 㔠 㔡 㔢 㔣 㔤 㔥 㔦 㔧 㔨 㔩 㔪 㔫 㔬 㔭 㔮 㔯 㔰 㔱 㔲 㔳 㔴 㔵 㔶 㔷 㔸 㔹 㔺 㔻 㔼 㔽 㔾 㔿 㕀 㕁 㕂 㕃 㕄 㕅 㕆 㕇 㕈 㕉 㕊 㕋 㕌 㕍 㕎 㕏 㕐 㕑 㕒 㕓 㕔 㕕 㕖 㕗 㕘 㕙 㕚 㕛 㕜 㕝 㕞 㕟 㕠 㕡 㕢 㕣 㕤 㕥 㕦 㕧 㕨 㕩 㕪 㕫 㕬 㕭 㕮 㕯 㕰 㕱 㕲 㕳 㕴 㕵 㕶 㕷 㕸 㕹 㕺 㕻 㕼 㕽 㕾 㕿 㖀 㖁 㖂 㖃 㖄 㖅 㖆 㖇 㖈 㖉 㖊 㖋 㖌 㖍 㖎 㖏 㖐 㖑 㖒 㖓 㖔 㖕 㖖 㖗 㖘 㖙 㖚 㖛 㖜 㖝 㖞 㖟 㖠 㖡 㖢 㖣 㖤 㖥 㖦 㖧 㖨 㖩 㖪 㖫 㖬 㖭 㖮 㖯 㖰 㖱 㖲 㖳 㖴 㖵 㖶 㖷 㖸 㖹 㖺 㖻 㖼 㖽 㖾 㖿 㗀 㗁 㗂 㗃 㗄 㗅 㗆 㗇 㗈 㗉 㗊 㗋 㗌 㗍 㗎 㗏 㗐 㗑 㗒 㗓 㗔 㗕 㗖 㗗 㗘 㗙 㗚 㗛 㗜 㗝 㗞 㗟 㗠 㗡 㗢 㗣 㗤 㗥 㗦 㗧 㗨 㗩 㗪 㗫 㗬 㗭 㗮 㗯 㗰 㗱 㗲 㗳 㗴 㗵 㗶 㗷 㗸 㗹 㗺 㗻 㗼 㗽 㗾 㗿 㘀 㘁 㘂 㘃 㘄 㘅 㘆 㘇 㘈 㘉 㘊 㘋 㘌 㘍 㘎 㘏 㘐 㘑 㘒 㘓 㘔 㘕 㘖 㘗 㘘 㘙 㘚 㘛 㘜 㘝 㘞 㘟 㘠 㘡 㘢 㘣 㘤 㘥 㘦 㘧 㘨 㘩 㘪 㘫 㘬 㘭 㘮 㘯 㘰 㘱 㘲 㘳 㘴 㘵 㘶 㘷 㘸 㘹 㘺 㘻 㘼 㘽 㘾 㘿 㙀 㙁 㙂 㙃 㙄 㙅 㙆 㙇 㙈 㙉 㙊 㙋 㙌 㙍 㙎 㙏 㙐 㙑 㙒 㙓 㙔 㙕 㙖 㙗 㙘 㙙 㙚 㙛 㙜 㙝 㙞 㙟 㙠 㙡 㙢 㙣 㙤 㙥 㙦 㙧 㙨 㙩 㙪 㙫 㙬 㙭 㙮 㙯 㙰 㙱 㙲 㙳 㙴 㙵 㙶 㙷 㙸 㙹 㙺 㙻 㙼 㙽 㙾 㙿 㚀 㚁 㚂 㚃 㚄 㚅 㚆 㚇 㚈 㚉 㚊 㚋 㚌 㚍 㚎 㚏 㚐 㚑 㚒 㚓 㚔 㚕 㚖 㚗 㚘 㚙 㚚 㚛 㚜 㚝 㚞 㚟 㚠 㚡 㚢 㚣 㚤 㚥 㚦 㚧 㚨 㚩 㚪 㚫 㚬 㚭 㚮 㚯 㚰 㚱 㚲 㚳 㚴 㚵 㚶 㚷 㚸 㚹 㚺 㚻 㚼 㚽 㚾 㚿 㜀 㜁 㜂 㜃 㜄 㜅 㜆 㜇 㜈 㜉 㜊 㜋 㜌 㜍 㜎 㜏 㜐 㜑 㜒 㜓 㜔 㜕 㜖 㜗 㜘 㜙 㜚 㜛 㜜 㜝 㜞 㜟 㜠 㜡 㜢 㜣 㜤 㜥 㜦 㜧 㜨 㜩 㜪 㜫 㜬 㜭 㜮 㜯 㜰 㜱 㜲 㜳 㜴 㜵 㜶 㜷 㜸 㜹 㜺 㜻 㜼 㜽 㜾 㜿 㝀 㝁 㝂 㝃 㝄 㝅 㝆 㝇 㝈 㝉 㝊 㝋 㝌 㝍 㝎 㝏 㝐 㝑 㝒 㝓 㝔 㝕 㝖 㝗 㝘 㝙 㝚 㝛 㝜 㝝 㝞 㝟 㝠 㝡 㝢 㝣 㝤 㝥 㝦 㝧 㝨 㝩 㝪 㝫 㝬 㝭 㝮 㝯 㝰 㝱 㝲 㝳 㝴 㝵 㝶 㝷 㝸 㝹 㝺 㝻 㝼 㝽 㝾 㝿 㞀 㞁 㞂 㞃 㞄 㞅 㞆 㞇 㞈 㞉 㞊 㞋 㞌 㞍 㞎 㞏 㞐 㞑 㞒 㞓 㞔 㞕 㞖 㞗 㞘 㞙 㞚 㞛 㞜 㞝 㞞 㞟 㞠 㞡 㞢 㞣 㞤 㞥 㞦 㞧 㞨 㞩 㞪 㞫 㞬 㞭 㞮 㞯 㞰 㞱 㞲 㞳 㞴 㞵 㞶 㞷 㞸 㞹 㞺 㞻 㞼 㞽 㞾 㞿 㟀 㟁 㟂 㟃 㟄 㟅 㟆 㟇 㟈 㟉 㟊 㟋 㟌

- **Step 1** — shop 表 (brand × brand_user × shop × brand_shop 4 表 · LIMIT %s+1 OFFSET %s has_more 表 DB-level pagination)
- **Step 2** — 每 20 shop_ids 表 orders GROUP BY → 140 表 (20 × 7) 表
- 每 4 表 (active / inactive / shop_deleted / brand_removed) — 每 表 0, 表
- stale 表 SQL VIEW (v_shop_crawling_status) 表 表 API · 表 ad-hoc · BI 表 表 表 (single source of truth)
- 表 / 表 表 daily upsert + 4 表 表 (SUCCESS / PLATFORM_FAILED / SCRAPER_FAILED / FAILED) — 表 表 表 read

□□ (□□ □□)

- $1,000 \times 7 = 7,000$ 個 個 → **140 個 (98% 個)** 個 個個個 個

- **IO 0** · p99 latency
- **VIEW** stale SQL
- 1500 + idempotent re-fetch

200 batch select · VIEW SSOT · 400 ADR

Spring — Spring Data JPA + QueryDSL · @Subselect SQL VIEW · @EntityGraph

N+1 · Cursor

[commerce-comment-platform-be](https://github.com/PreAgile/commerce-comment-platform-be))

50

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- **Single-Flight Coordinator** — Hexagonal Architecture (Port-Adapter) + Decorator 50. N → 7000 spec 80+ · 22 new unit spec.
- **SafeCron** — cron NestJS DI (setModuleRef) + Slack 14.
- 1,000+ SELECT FOR UPDATE + (@Transactional). row
- 200+ (1 ≈ 1GB+ RAM). ASN 32 / 512 GB RAM 8. Traefik + 5-step
- AI → TOCTOU 4 → API Key/JWT + Layered Idempotency
- 6 (Shannon · zlib · n-gram · run) + 가 Bell + Polarity Amplification + tanh box) + OR

JVM/Spring — Node.js Java/Kotlin/Spring

50 Node.js / NestJS ADR (Architecture Decision Record — 1000 1000 1000)

NestJS Java/Kotlin/Spring/JPA ADR

ADR 49 · 47 · 17

[commerce-comment-platform-be](https://github.com/PreAgile/commerce-comment-platform-be) — Java / Spring Boot / JPA

- Stage 0 (raw JDBC) → Stage 5 (DDD / MSA) · Lombok · record / sealed · OSIV=false
- **ADR-006** 400 — idempotency_key + 4 + Redis SETNX 1 + DB UNIQUE
- **ADR-004** — Redisson watchdog vs SET NX+Lua
- **ADR-BE-007** — REPEATABLE READ + RC (EXP-06 4 phantom)

- **ADR-BE-008** **Cursor** — **Cursor** **row constructor** **Cursor** (EXP-09b 9 **Cursor** **Cursor**)
- **ADR-BE-009 No-Offset Pagination** — **Cursor** **row constructor** **Cursor** (EXP-07 1M OFFSET 171ms → **Cursor**)
- **Cursor** — EXP-02 4 GET_LOCK **Cursor**, EXP-03 @Version **Cursor** silent Lost Update 74 / OptLockEx 86 **Cursor**, EXP-04 N+1 4-depth 121 prepStmt → JOIN FETCH 1, EXP-13 dirty checking 132x readOnly **Cursor**, EXP-14 IDENTITY=saveAll batch **Cursor**

[commerce-batch-orchestrator](https://github.com/PreAgile/commerce-batch-orchestrator) — **Cursor** / **Kafka** / **Outbox**

- **ADR-005 Outbox** **Cursor** → **Debezium CDC** **Cursor** — Dual-Write **Cursor**
- **ADR-MQ-006 RabbitMQ vs Kafka** **Cursor** **Cursor** — RabbitMQ classic queue replay 가 · HOL blocking · Consumer Group **Cursor** **Cursor** **Cursor** **Cursor** **Cursor** + RabbitMQ **Cursor**
- **ADR-MQ-007 Spring Batch Reader 4** **Cursor** — QuerydslZeroOffsetItemReader **Cursor**
- **ADR-MQ-008 Consumer** **Cursor** + **DLQ** + **retry topic** — At-least-once + **Cursor** = Effectively-Once
- **ADR-MQ-009 CooperativeStickyAssignor (KIP-429)** — Rebalance stop-the-world **Cursor**
- **Cursor** **Cursor** — E-MQ-01 Direct publish **Cursor** → E-MQ-02 Outbox **Cursor** → E-MQ-03 Debezium CDC, E-MQ-04~07 Reader 4 100 row **Cursor**, E-MQ-08/09 RabbitMQ → Kafka throughput, E-MQ-10 Eager vs Cooperative Sticky rebalance **Cursor**

Cursor **RabbitMQ** **Cursor** **Cursor** **Cursor** **ADR-MQ-006** **Cursor** **Cursor** **Cursor**.

[commerce-external-gateway-kt](https://github.com/PreAgile/commerce-external-gateway-kt) — **Cursor** **Cursor** (**Kotlin** / **Coroutines** / **Resilience4j**)

- **ADR-002 Coroutines vs Virtual Threads** — Coroutines **Cursor** (**Cursor** **Cursor**)
- **ADR-007 Circuit Breaker** **Cursor** — 50% / 70% / 90% **Cursor** 70% **Cursor** (CB-1/2/3 **Cursor**)
- **ADR-SCR-008 Single-Flight Coordinator** — TS **Cursor** 7 **Cursor** **Cursor** Kotlin Coroutines Mutex / Deferred **Cursor**. SF-2 99.9% coalescing **Cursor**
- **ADR-SCR-009 3-Layer** **Cursor** **Cursor** — Memory L1 → Redis L2 → **Cursor** **Cursor** L3 (NAVER **Cursor** **Cursor** **Cursor**)
- **ADR-SCR-010 Bulkhead** **Cursor** — Semaphore vs FixedThreadPool **Cursor** **Cursor** **Cursor**
- **ADR-SCR-011 Resilience4j** **Cursor** **Cursor** — Bulkhead → CB → Retry → TimeLimiter → RateLimiter (Retry **Cursor** CB **Cursor** CB **Cursor** N **Cursor** **Cursor**)
- **Cursor** **Cursor** — EXP-WC-01 WebClient .block() EventLoop pinning **Cursor** 1/10 **Cursor** **Cursor** → **Cursor** **Cursor** **Cursor**

Cursor — **Cursor**(I-BRICKS) | | **Cursor** 2021.05 ~ 2024.11

Cursor **Cursor** **Cursor** **Cursor** **Cursor** **Cursor**, **Cursor** **Cursor**, **Cursor** **Cursor**.

- **EBS** **Cursor** **Cursor** **Cursor** (**Kafka** · **Apache Nifi** · **Elasticsearch**) — **Cursor** **Cursor** **Cursor** Kafka producer / consumer + Nifi flow + Elasticsearch **Cursor** **Cursor**. **Cursor** **Cursor** INSERT **Cursor** **Cursor** → Kafka **Cursor** **Cursor** + Nifi **Cursor** **Cursor** 2 **Cursor** **Cursor** **Cursor**, **Cursor** **Cursor** **Cursor** 50% **Cursor**. (Kafka **Cursor** **Cursor** **Cursor** — JVM **Cursor** **Cursor** **Cursor** commerce-batch-orchestrator **Cursor** **Cursor** **Cursor**)

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